Atomo exercises

Exercise on Atoms, Isotopes and Ions

Complete the following table. Minimal data are given in each case. You must deduce the remaining quantities.

Remember

Remember:
- Z = atomic number = number of protons
- A = mass number = protons + neutrons
- Q = charge = protons - electrons

Element name	Symbol (with Z, A and Q)	Z	A	Q	N° protons	N° electrons	N° neutrons
Hydrogen		1	1	0			
	${}^{4}_{2}He^{0}$						
Lithium			7	+1	3		
	${}^{12}_{6}C^{0}$						
Oxygen		8		-2			8
	${}^{23}_{11}Na^{+}$						
Chlorine					17	18	18
	${}^{40}_{20}Ca^{2+}$						
Iron		26				23	30
Uranium	${}^{238}_{92}U^{0}$						

Solutions exercise 1

Element name	Symbol (with Z, A and Q)	Z	A	Q	N° protons	N° electrons	N° neutrons
Hydrogen	${}^{1}_{1}H^{0}$	1	1	0	1	1	0
Helium	${}^{4}_{2}He^{0}$	2	4	0	2	2	2
Lithium	${}^{7}_{3}Li^{+}$	3	7	+1	3	2	4
Carbon	${}^{12}_{6}C^{0}$	6	12	0	6	6	6
Oxygen	${}^{16}_{8}O^{2-}$	8	16	-2	8	10	8
Sodium	${}^{23}_{11}Na^{+}$	11	23	+1	11	10	12
Chlorine	${}^{35}_{17}Cl^{-}$	17	35	-1	17	18	18
Calcium	${}^{40}_{20}Ca^{2+}$	20	40	+2	20	18	20
Iron	${}^{56}_{26}Fe^{3+}$	26	56	+3	26	23	30
Uranium	${}^{238}_{92}U^{0}$	92	238	0	92	92	146

Extension: applications of some isotopes and ions

1. Carbon: -Carbon-12 (^{12}C): It is used as the reference to define atomic mass. The entire atomic mass scale is based on this isotope. -Carbon-14 (^{14}C): used to determine the age of organic materials (bones, wood, plants, tissues) up to approximately 50,000–60,000 years old. It is a fundamental technique in archaeology, geology and paleontology because unstable ^{14}C decays at a constant rate after the organism's death.
2. Oxygen (oxide ion, O^{2-}): Very important in the formation of compounds such as water or metal oxides. It is essential in biological and geological processes.
3. Sodium (Na^{+}): Essential for the functioning of the nervous system. It enables the transmission of nerve impulses.
4. Calcium (Ca^{2+}): Key in the formation of bones and teeth. It also plays a role in muscle contraction.
5. Uranium-238 (^{238}U): It is used as fuel in nuclear power plants and in the dating of very old rocks.

Competency-based exercises

1. A calcium ion has a charge of +2 and 20 protons. How many electrons does it have? Explain what this charge means.
2. Chlorine usually forms an ion with a charge of -1. If it has Z = 17, calculate the number of electrons and explain why it gains electrons.
3. An atom has 11 protons, 12 neutrons and 10 electrons. Identify the element, its charge and write its full symbol.
4. In an isotonic drink there are sodium ions (Na^{+}). Explain why they are important for the body after exercising.
5. A neutral atom has A = 40 and Z = 20. Calculate protons, electrons and neutrons. Which element is it?
6. Iron can form the ion Fe^{3+}. Explain how many electrons it has lost compared to the neutral atom and why metals tend to lose electrons.

Solutions competency exercises

1. It has 18 electrons. It has lost 2 electrons, which is why its charge is positive.
2. It has 18 electrons. It gains one electron to complete its outer shell and become more stable.
3. It is sodium ($^{23}_{11}Na^{+}$). It has a +1 charge because it has lost one electron.
4. They help replenish mineral salts lost through sweat and allow proper muscular and nervous function.
5. Protons = 20, electrons = 20, neutrons = 20. It is calcium.
6. It has lost 3 electrons. Metals tend to lose electrons because this allows them to reach a more stable electronic configuration.